

Lecture 5: Activity Diagrams

IT2030 – Software Engineering

Lesson Learning Outcomes

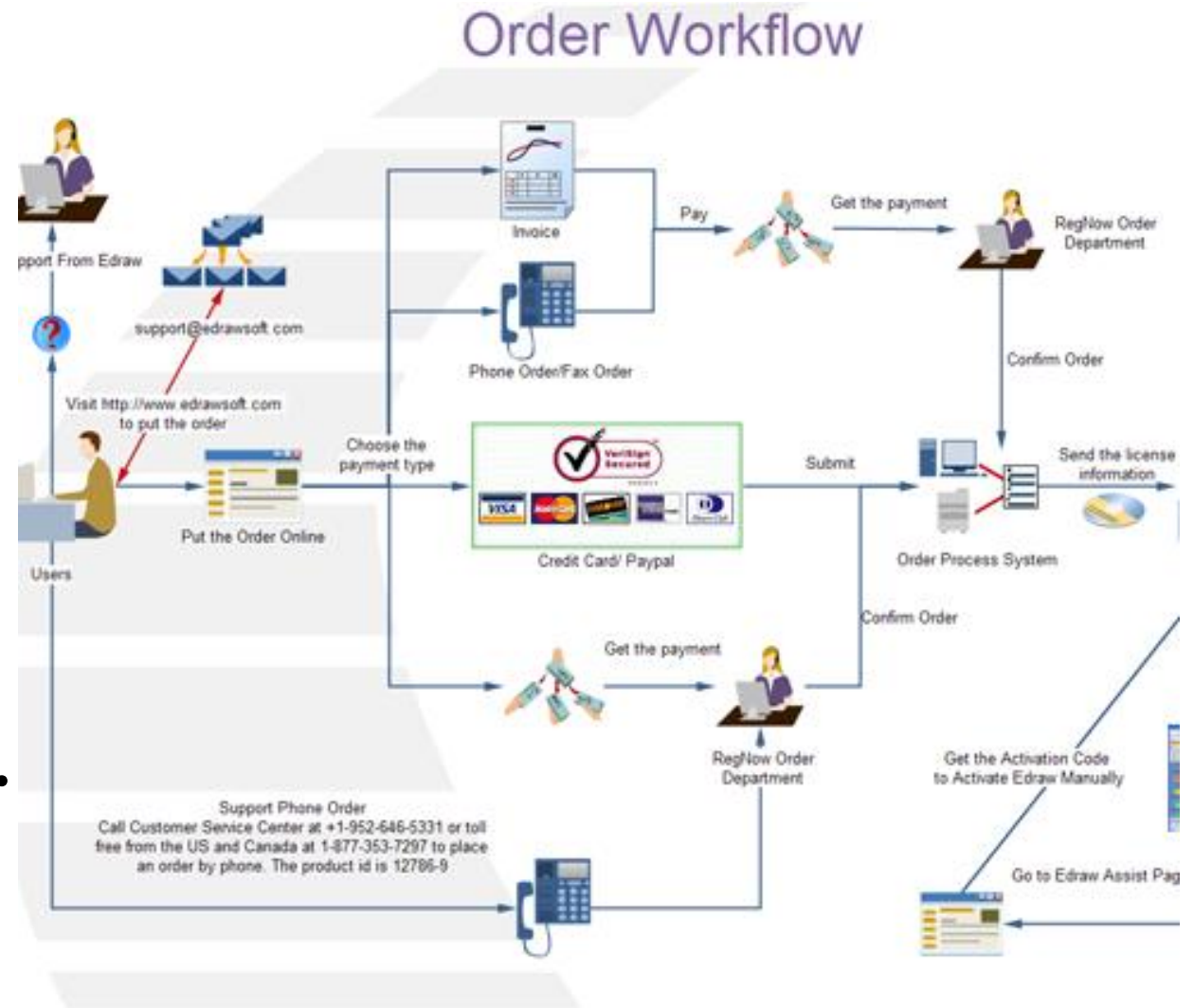
- Define activity diagrams and explain their purpose in modeling workflows and system behavior.
- Identify the main elements of an activity diagram, including actions, transitions, decision/merge nodes, forks/joins, and swimlanes.
- Differentiate between branching, merging, forking, and joining in activity diagrams.
- Interpret activity diagrams to describe the flow of activities for a given business or system process.
- Create accurate activity diagrams from textual scenarios, applying correct UML notation.

Content

- Introduction.
- Elements of Activity diagram.
- Partitioning an activity diagram.
- Applying activity diagrams in real world applications.

What is an Activity Diagram?

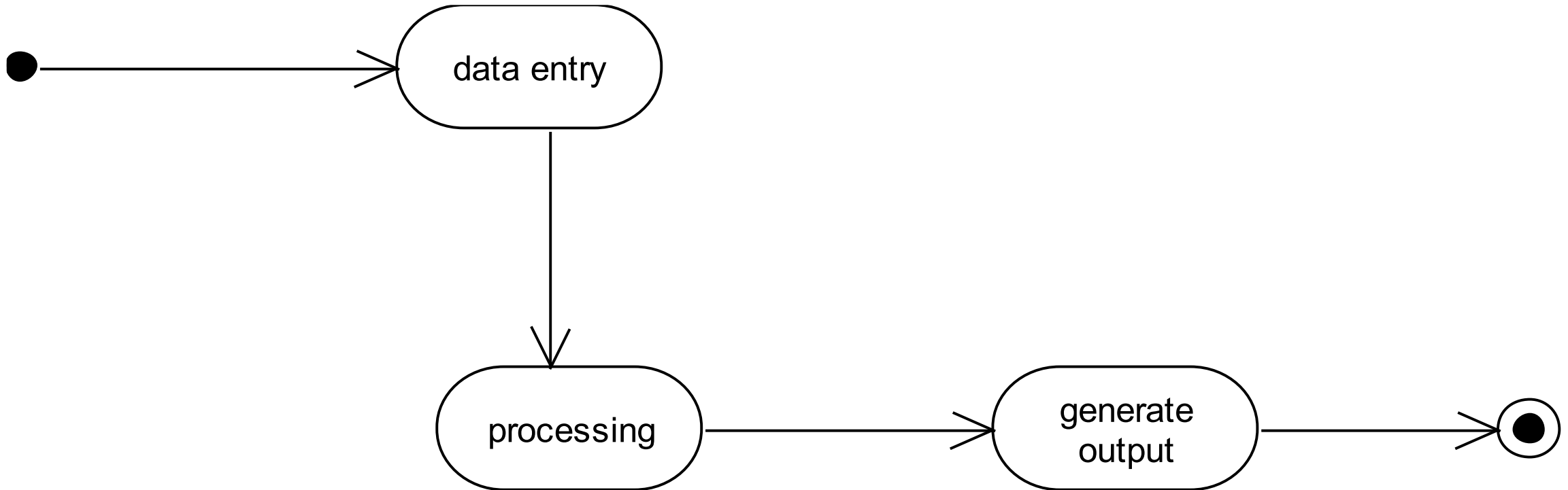
- An Activity Diagram models
 - Activities of a system.
 - Dependencies between activities.
 - Represents Workflows.



When to Use Activity Diagrams ?

- Modeling business processes.
- Analyzing system functionality to identify the use cases.
- Analyzing individual use cases in detail.
- Clarifying concurrency issues.

A simple Activity Diagram



Elements of an Activity Diagram

❖ Action

- Simple Action
- Call Action

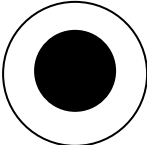
❖ Transition

❖ Controls

- Initial/Start
- Final/End
- Nodes
 - ✧ Decision/Branch
 - ✧ Merge
 - ✧ Fork
 - ✧ Join

Initial & Final Nodes

Represent the beginning and end of a diagram respectively.

- Initial/Start Node – Filled Circle 
- Final node is within circle 

Action Node

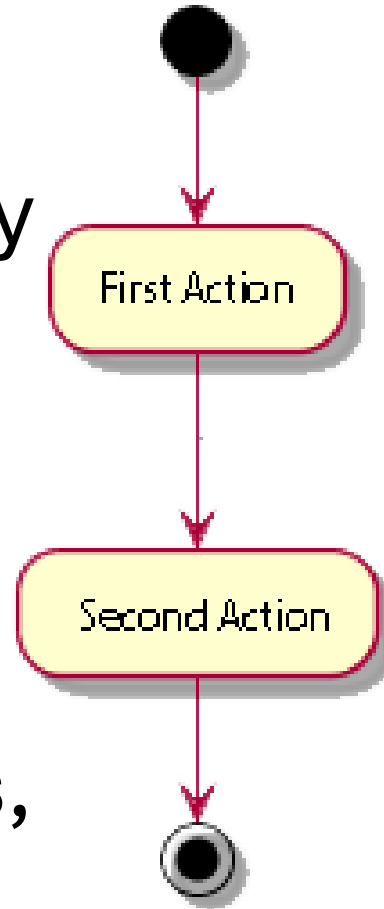
- An action is some task which needs to be done.
- Each action can be followed by another action .
- Represented with a rounded rectangle.
- Text in the action box should represent an activity (verb phrase in present tense).
- An activity is a sequence of actions.



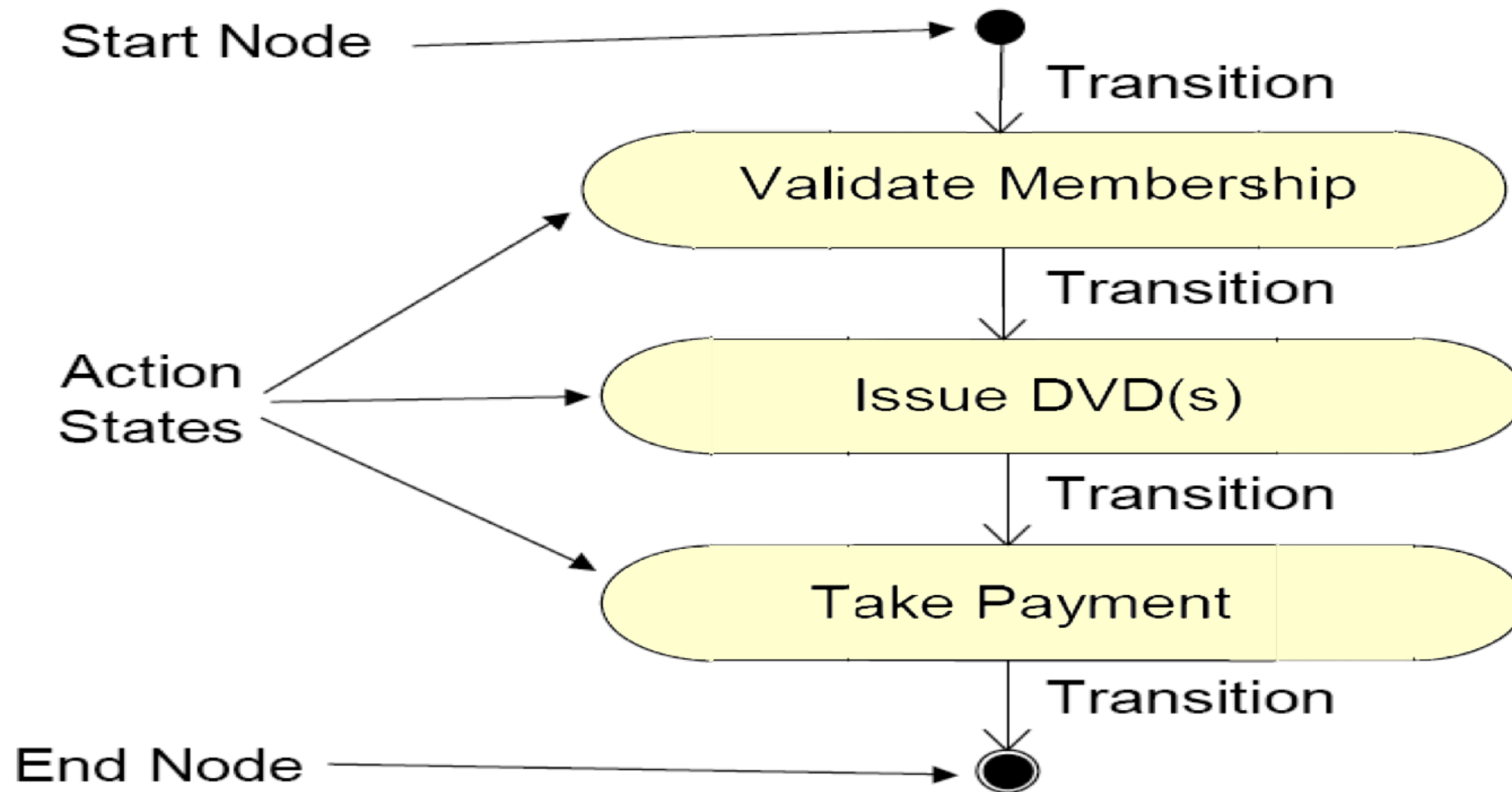
Transition



- Activity/Action nodes are connected by Transitions.
- Also known as a control flow/directed flow /edge
- When the execution of the node completes, execution proceeds to the node found on the output flow.



Activity Diagrams: Example



Activity - 1

Draw an activity diagram illustrating the process of becoming a registered member of the CA Sri Lanka Learning Management System.

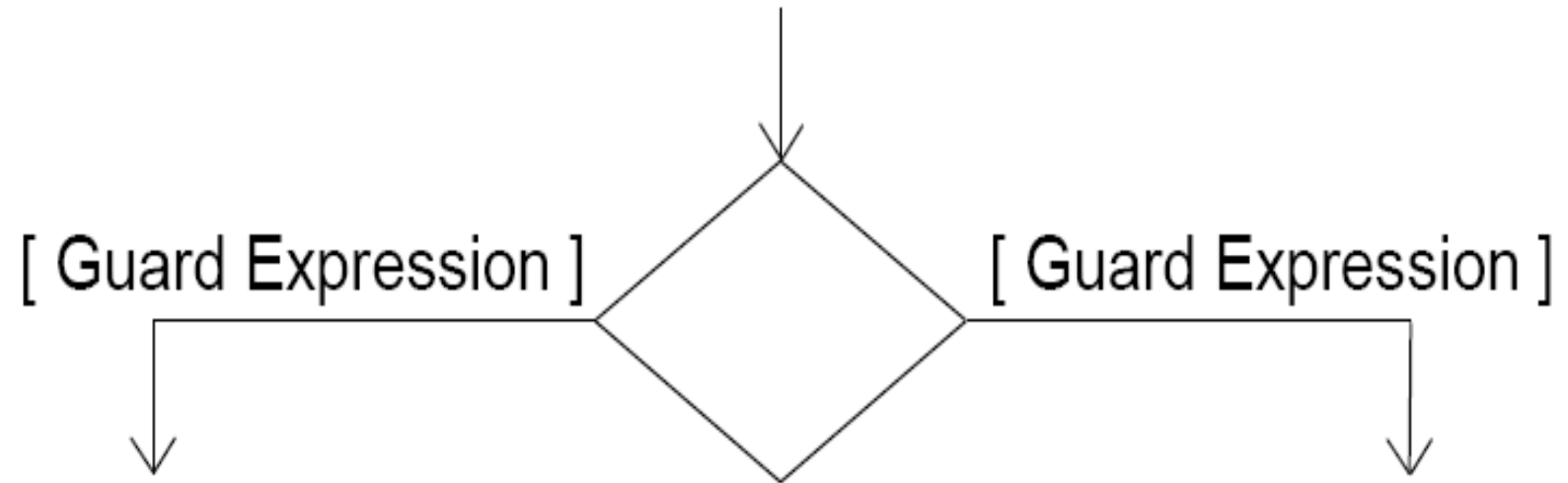
- To become a registered member, a student must first upload an image of their National Identity Card (NIC).
- The student must then complete the required registration forms and pay a deposit of Rs. 3,000.
- Once these steps are completed, the system displays the student's ID on the screen.

Decision Node

- A Decision represents a conditional flow of control:
 - The alternatives are mutually exclusive.
- Similar to IF/ELSE statements in programming languages like C/C++/Java.

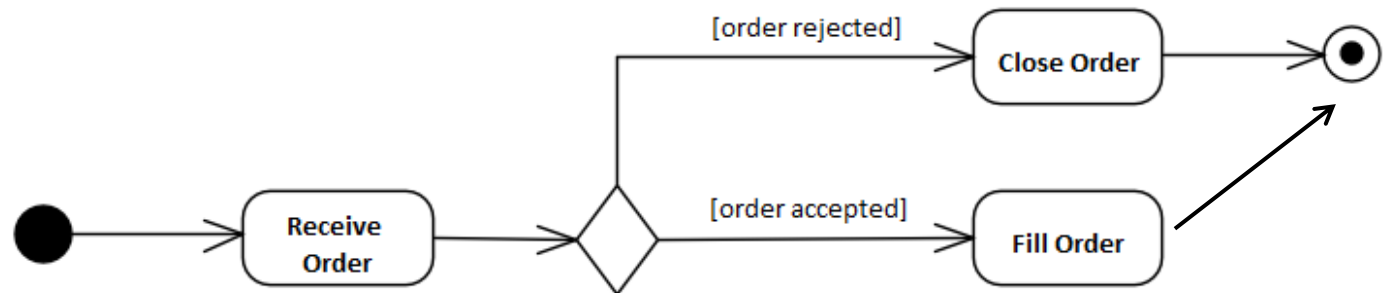
Decision Node

- A Decision/Branch is represented by:
 - a diamond at the branch point.
 - a guard expression for each possibility.

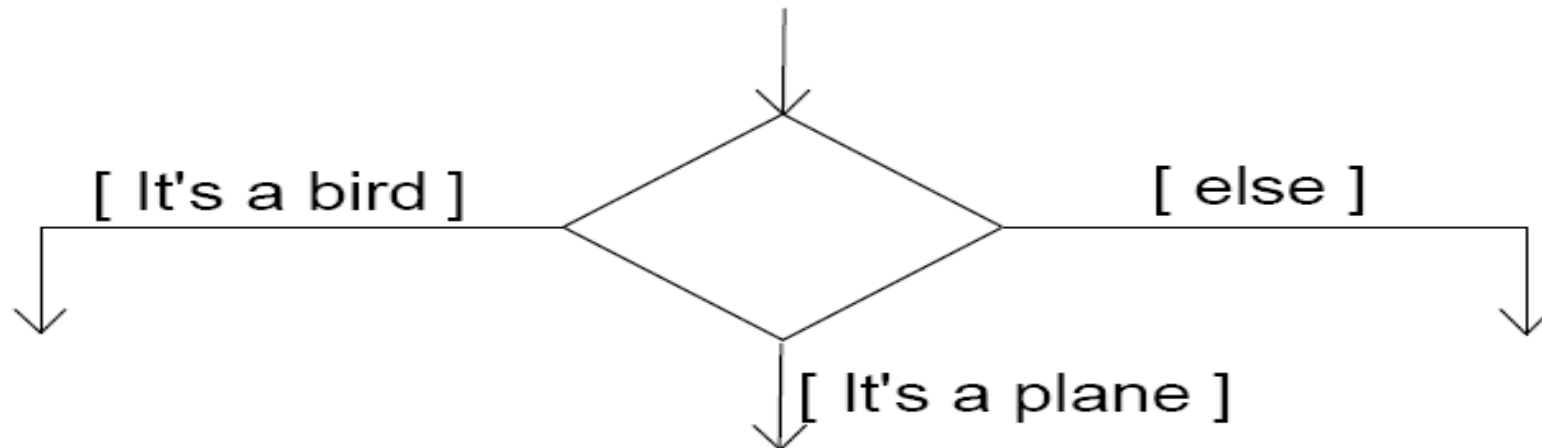
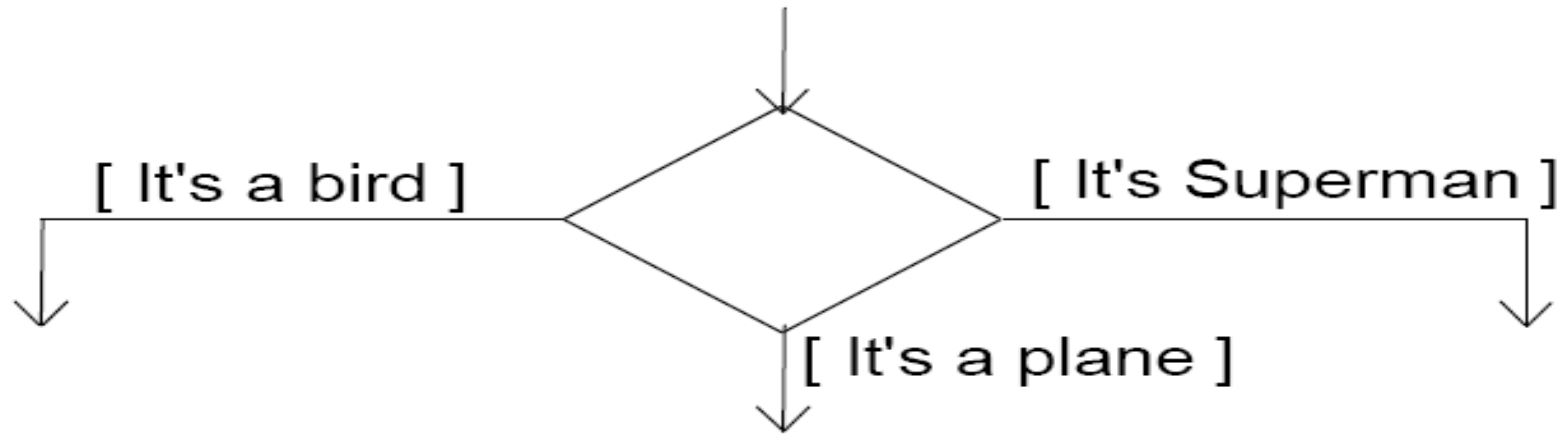


Decision Node

- Decision/Branch points.
 - Each branch must have a guard condition.
 - The flow of control flows down the single path where the branch condition is true.
 - There is no limit on the number of branches each branch point may have.

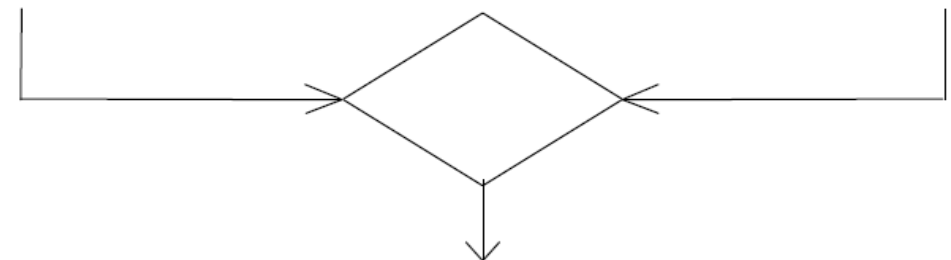


Multiple Branches: Examples

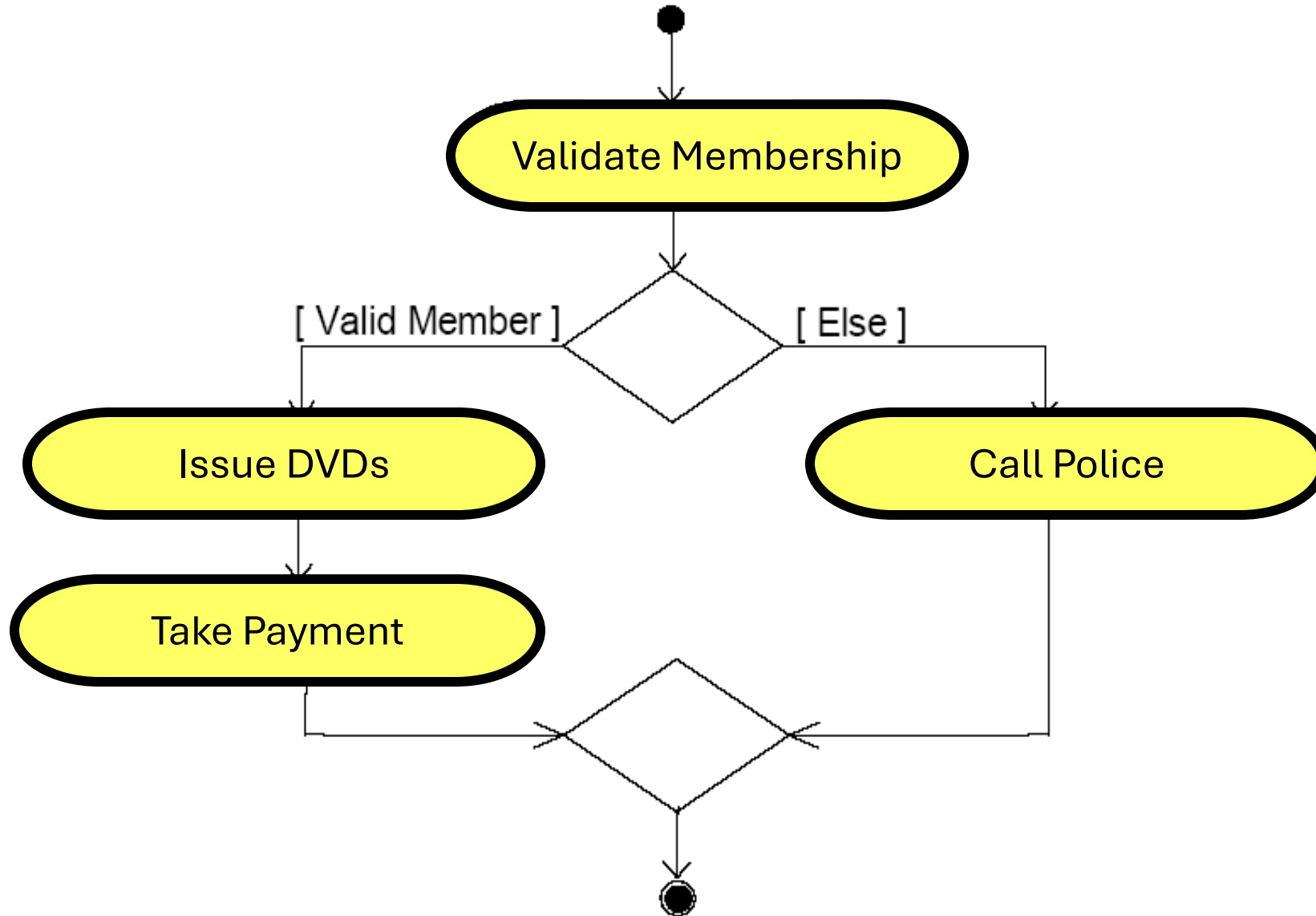


Merge Node

- A merge point is used to **merge the flow of control from two or more branch points back together.**
- The UML equivalent of ENDIF in pseudo code, or “}” in C/C++/Java.
- The diagram below shows a merge graphically – the diamond is optional:



Example





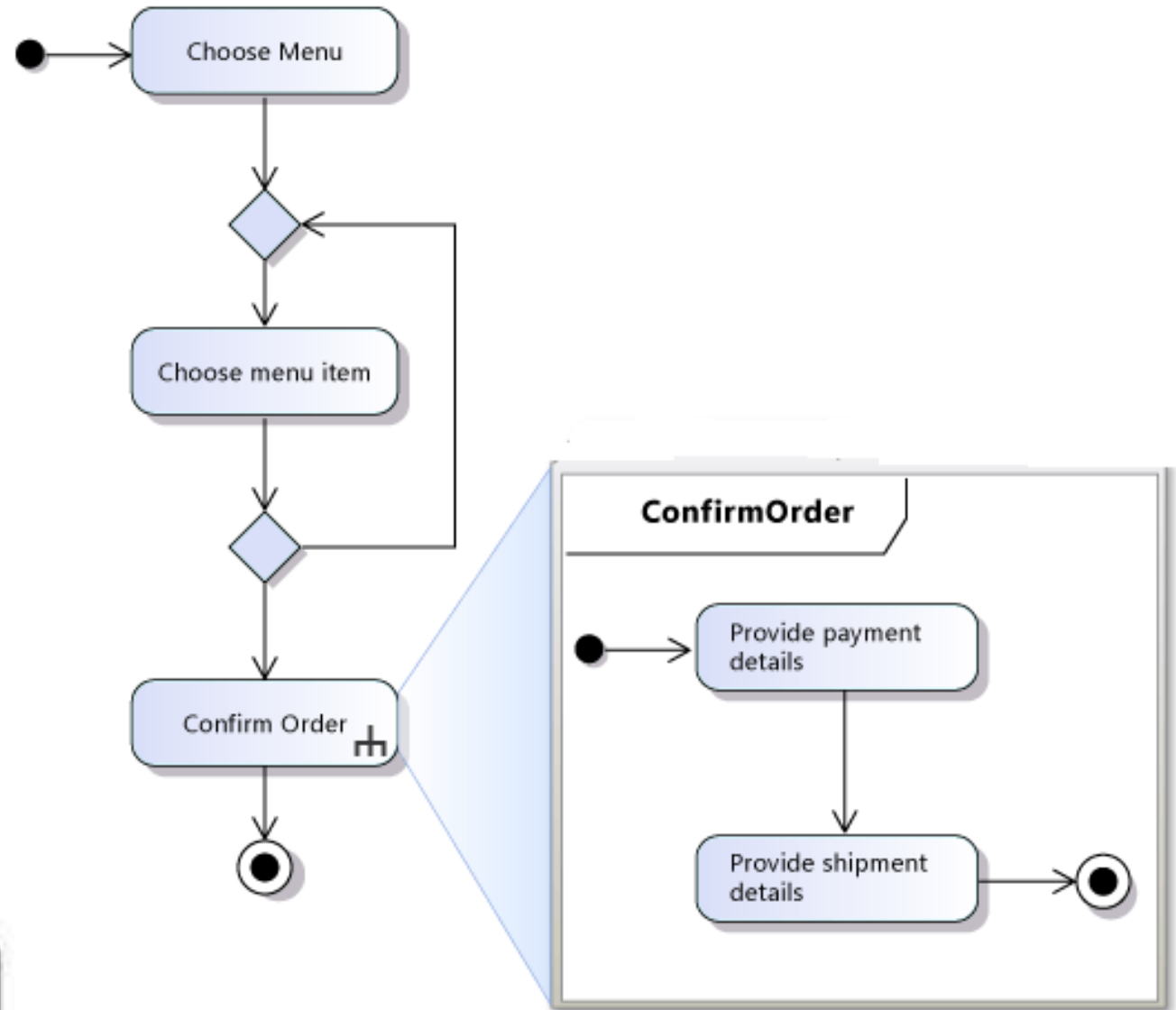
Activity - 2

Flight Booking with Qatar Airways

- To book a ticket through the Qatar Airways website, a guest must first register as a member. After registering, the member must select either Economy Class or Business Class.
- If Economy Class is selected, the member must check the baggage allowance. The default baggage allowance is 30 kg. If the member requires more baggage, a deposit must be paid for the additional baggage. After confirming the default baggage allowance or paying the required deposit, the member can proceed with the booking.
- If Business Class is selected, the member can proceed directly to the next step, as it includes a default baggage allowance of 40 kg.
- In the booking process, members must first complete the mandatory information form and then select a travel date. If a flight is available on the selected date, the member can book the ticket. If a flight is unavailable, the member must cancel the booking, and the process ends.

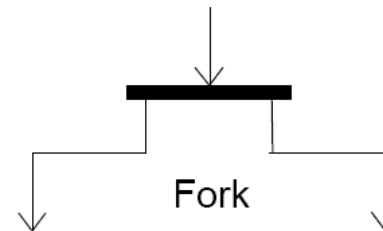
Call action /Sub Activity

- **Sub activity** is an activity that is defined in more detail on another activity diagram.
- It is indicated by a rake-style symbol within the action symbol.



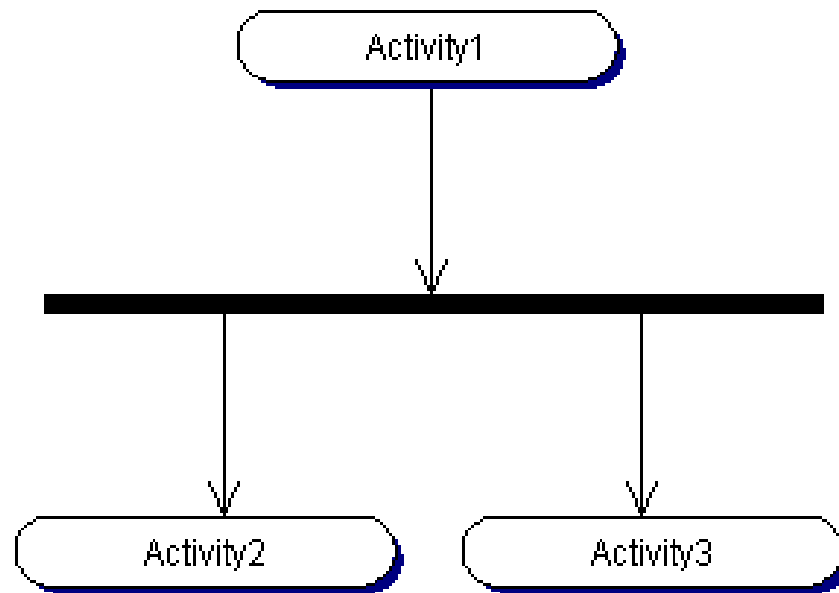
Forks & Joins

- Forks and joins are used to show activities that can occur at the **same time (in parallel)**.
 - This does not mean that the activities must occur concurrently in the finished software system.
 - It means that the order of execution of the activities can be whatever is convenient for the implementation.



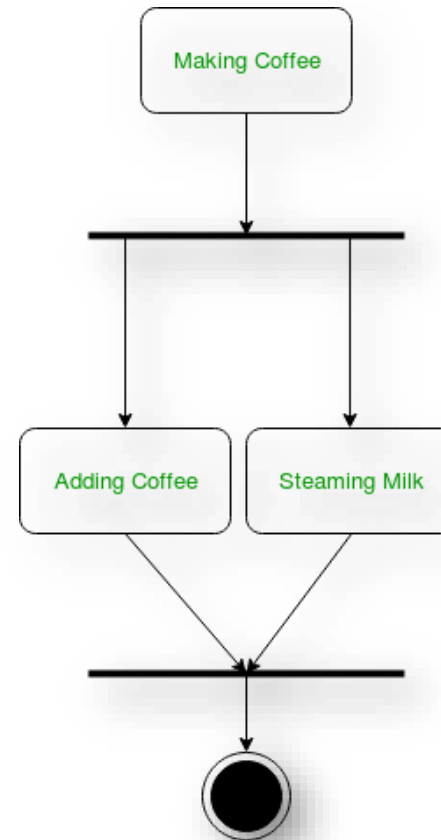
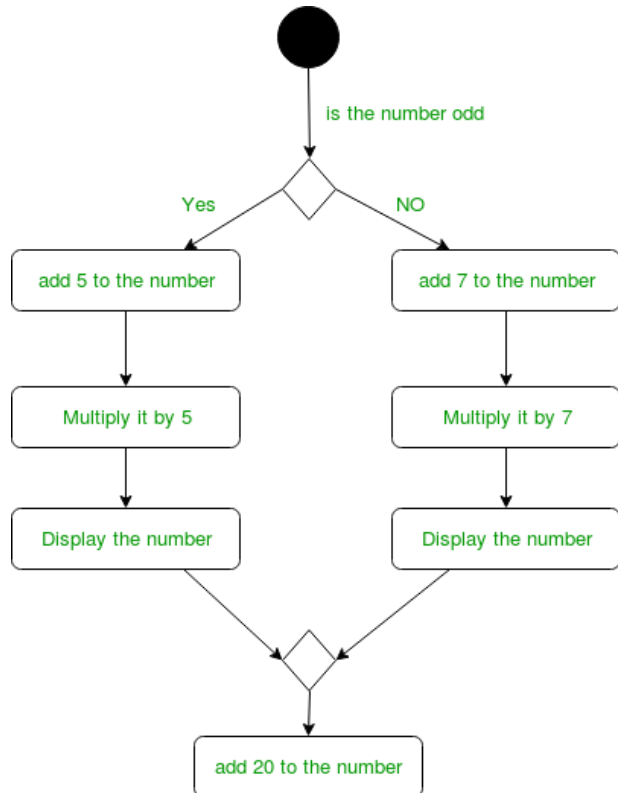
Fork

- A fork is when a single flow of control splits into two or more parallel (concurrent) flows of control.
- Represents a split in the flow of control.



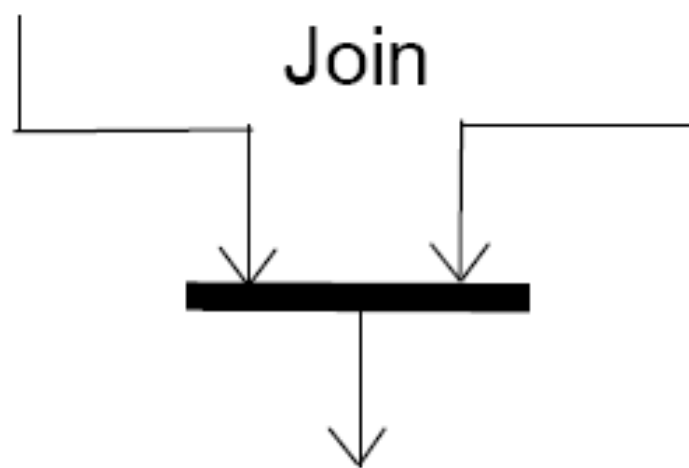
Fork

- Unlike a branch point, the control flows down **all forked paths**.
- With a branch point, the control flows down only **one path**.



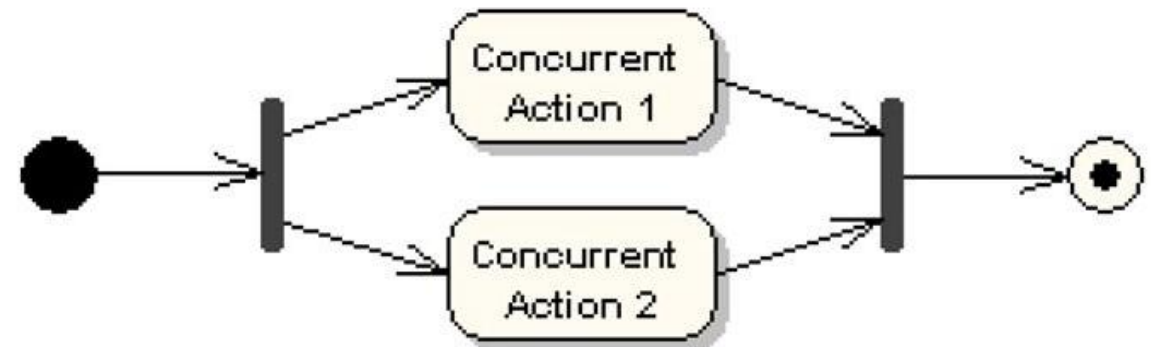
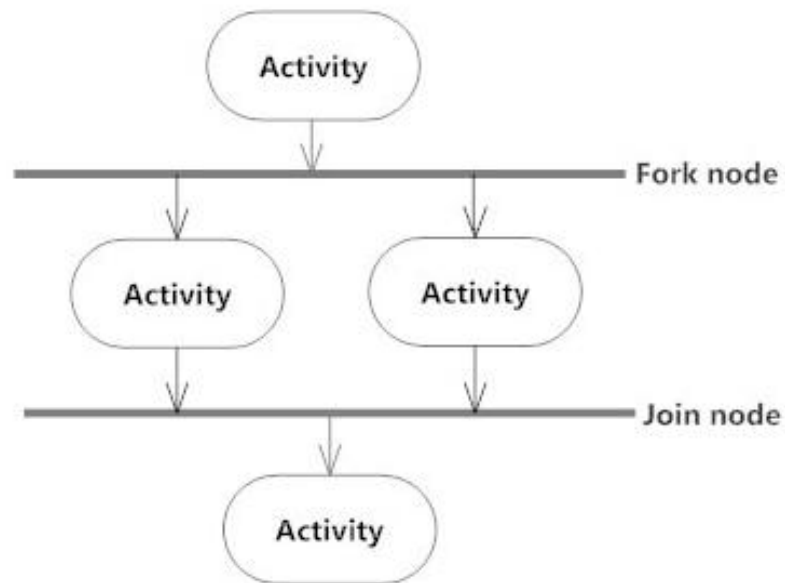
Join

- A join is when two or more flows of control merge into a single flow of control.
- Represents the merging of multiple flows of control back into one.
- **Every fork must have a join associated with it.**

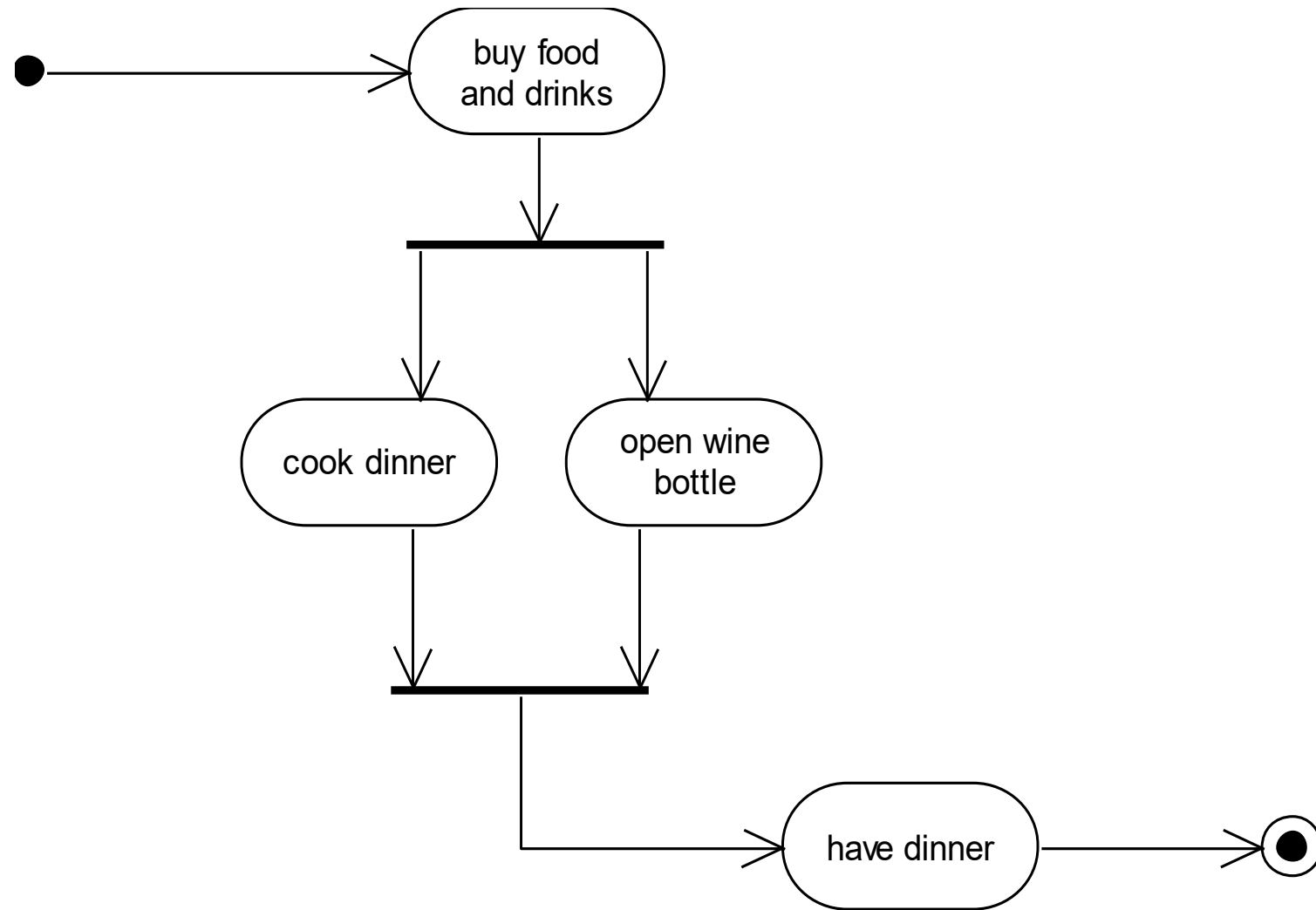


Join

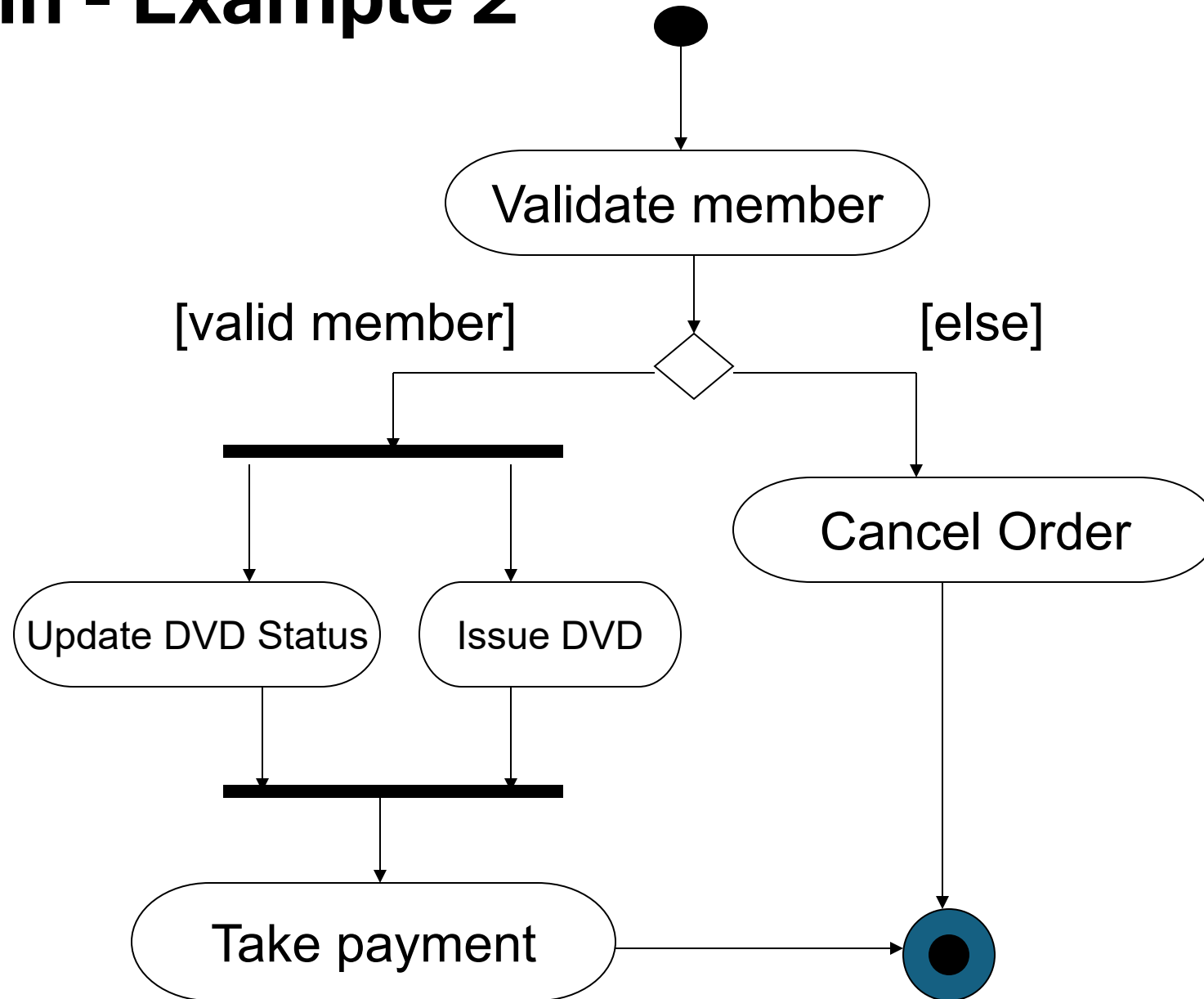
- A flow of control is also known as a thread.
- A synchronization bar is used to model forking and joining and is modeled as a thick horizontal or vertical bar.



Fork and Join - Example 1



Fork and Join - Example 2



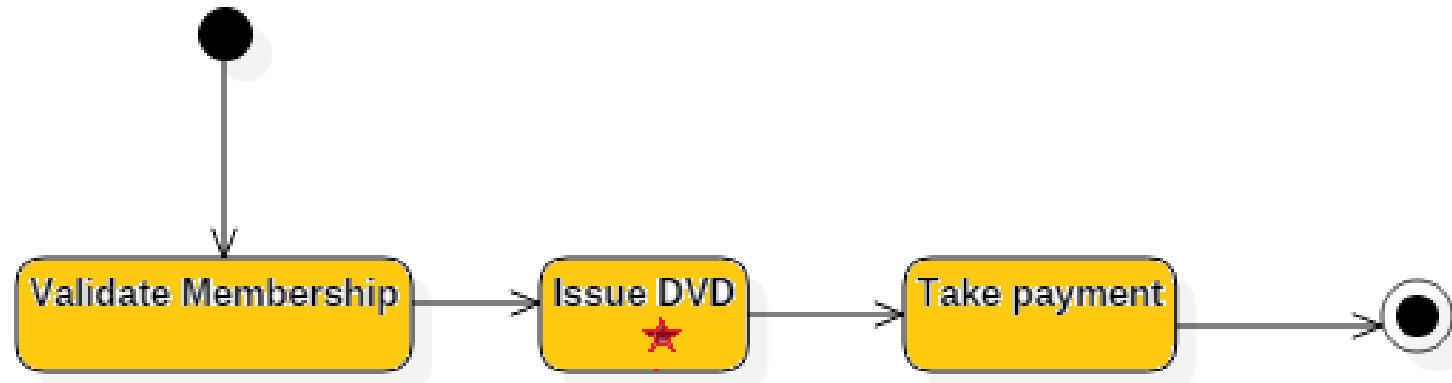
Activity - 3

Enrolling in a University

- Candidates who wish to enroll in the university must first complete and submit the required enrolment form.
- If the form has been completed incorrectly, the application will be rejected and the process will end. If the form has been completed correctly, the candidate will be enrolled in the university.
- After enrolment, the student must complete two activities: register for the seminar and make the required payment and attend the university overview presentation.
- These two activities may be completed concurrently. Once both activities have been completed, the student can begin attending classes.

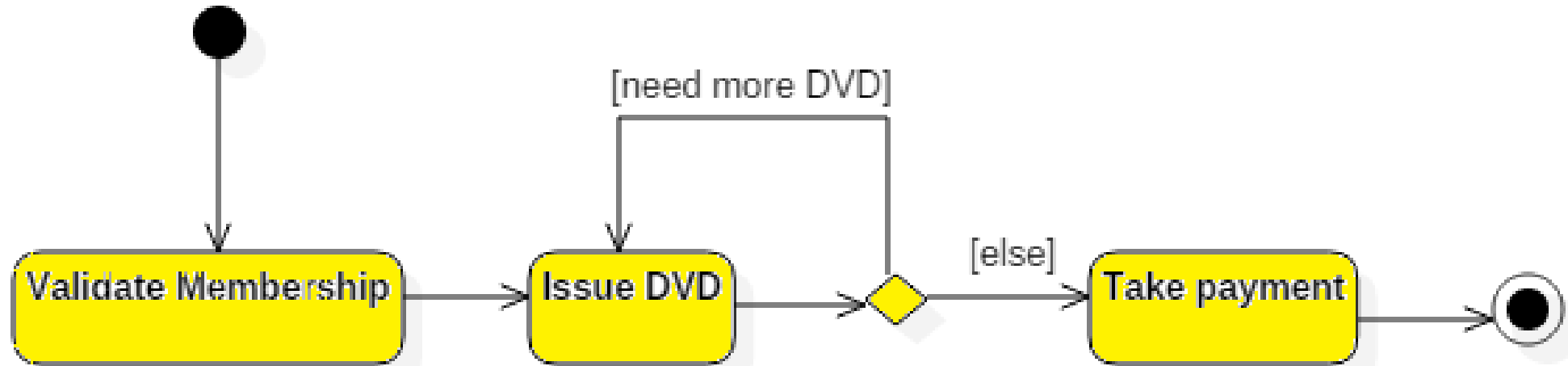
Iteration ★

- An **asterisk** inside an action state indicates that it may need to be **performed more than once**.
 - This notation is used to show iteration, without the unnecessary details of a loop construct.
- The next action state does not occur until the loop is finished.



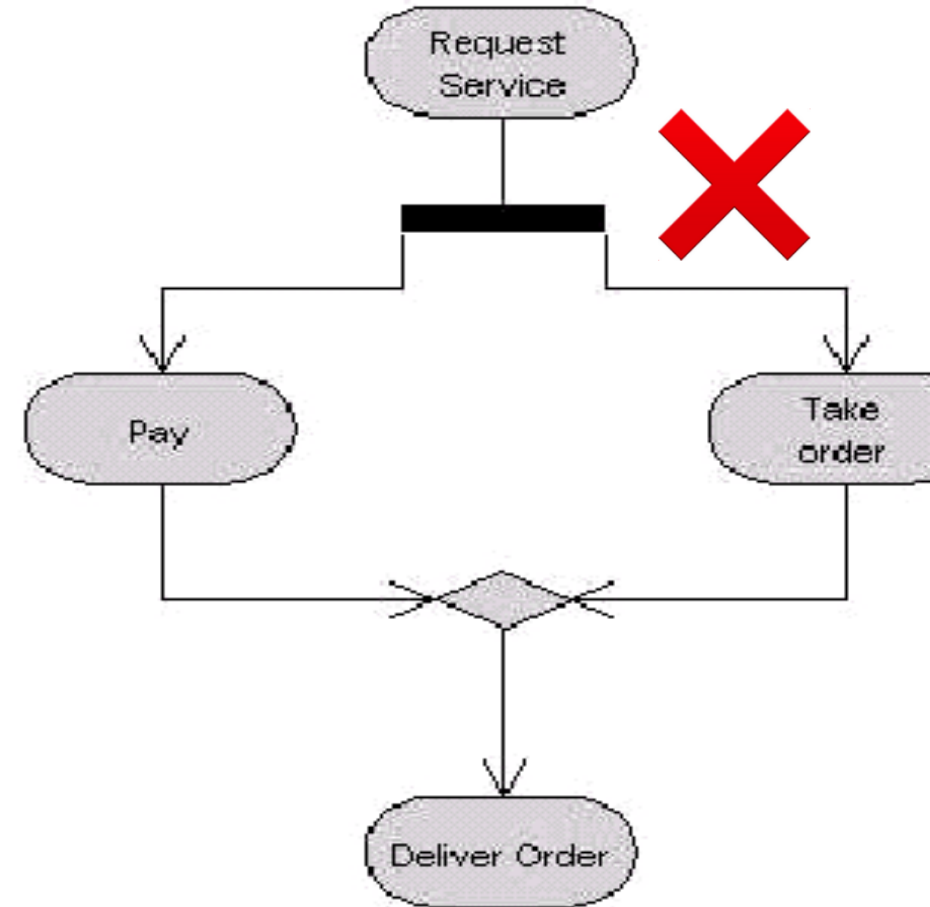
Iteration – Alternative Method

- Use of the **asterisk** to represent an iteration will not highlight the **loop termination conditions** and **number of repetitions**.
- Therefore, use of a **Decision Node** to represent an iteration would be ideal.



Common Mistakes

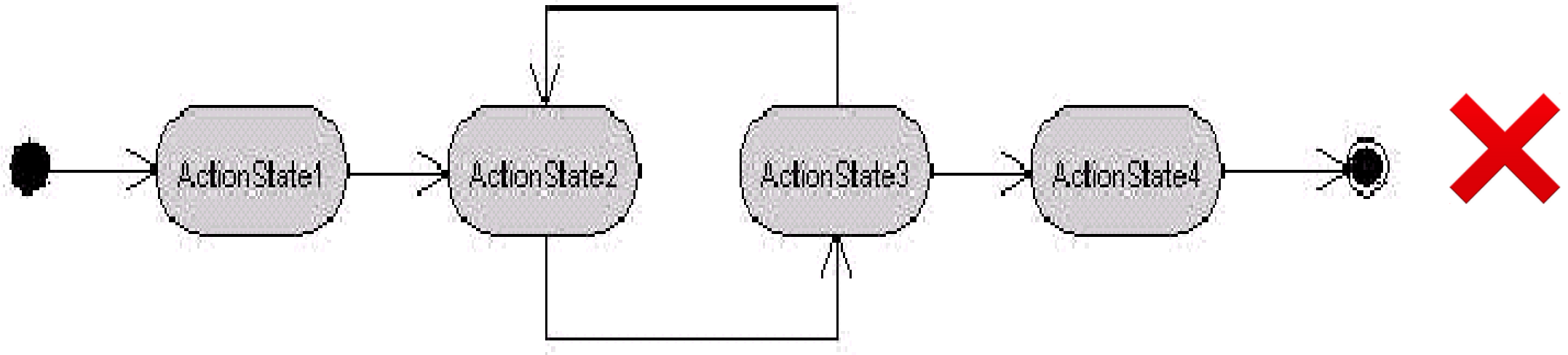
- Incorrect use of **forks, branch points, merge points and joins**.
- The fork means that the flow of control goes down both paths.
- The merge point indicates a merging of divergent paths, not flows.



Common Mistakes

Loops

- Guard conditions should be mentioned



Activity - 4

Developing a Software System for a Client

- The software development process begins when the Project Manager (PM) gathers the client's requirements.
- Based on these requirements, the UI Engineer (UIE) develops a prototype, which the Project Manager presents to the client.
- If the client is not satisfied with the prototype, the UI Engineer modifies it, and the Project Manager presents it to the client again.
- This process continues until the client is satisfied and approve the prototype.
- After the prototype is approved, two activities are carried out concurrently: the UI Engineer develops the user-interface screens, while the Software Engineer (SE) develops the software system.
- Once both activities have been completed, the Software Engineer integrates the user interface with the system.
- Finally, the Project Manager delivers the completed system to the client.
- The client signs off on the delivery, and the software development process ends.

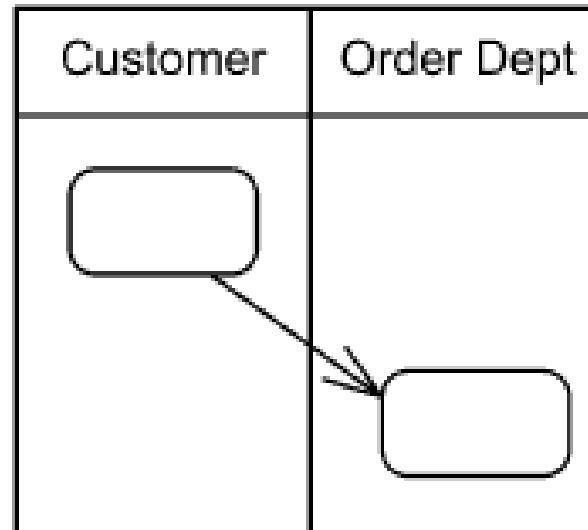
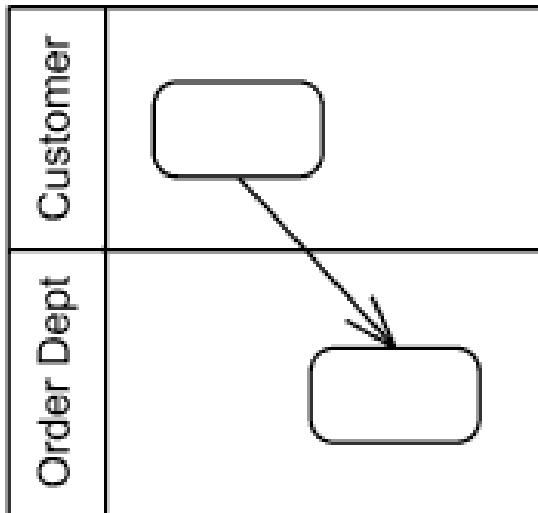
Partitioning

- An activity **partition** is an **activity group** for actions that have some common characteristic.
- Partitions often correspond to **organizational units** or **business actors** in a **business model**.

Partitions: SwimLanes

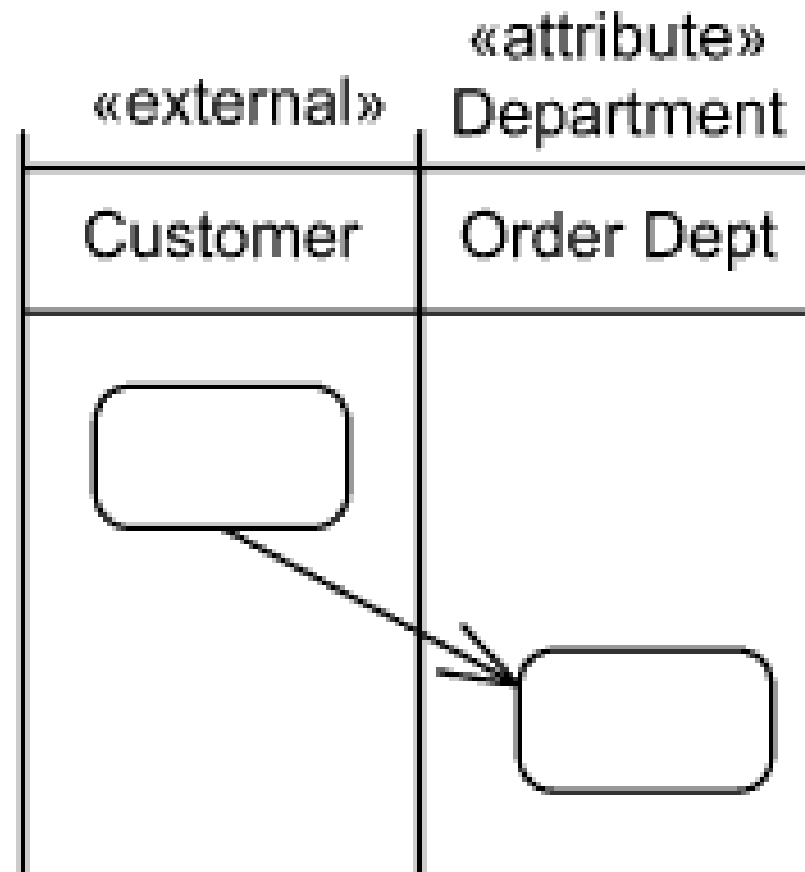
Activity **partition** may be shown using a **SwimLane** notation:

- with two, usually parallel lines, either horizontal or vertical
- a name labeling the partition in a box at one end.



Sub Partitioning in SwimLanes

- Order department is a subclass of the Department class.



Activity - 5

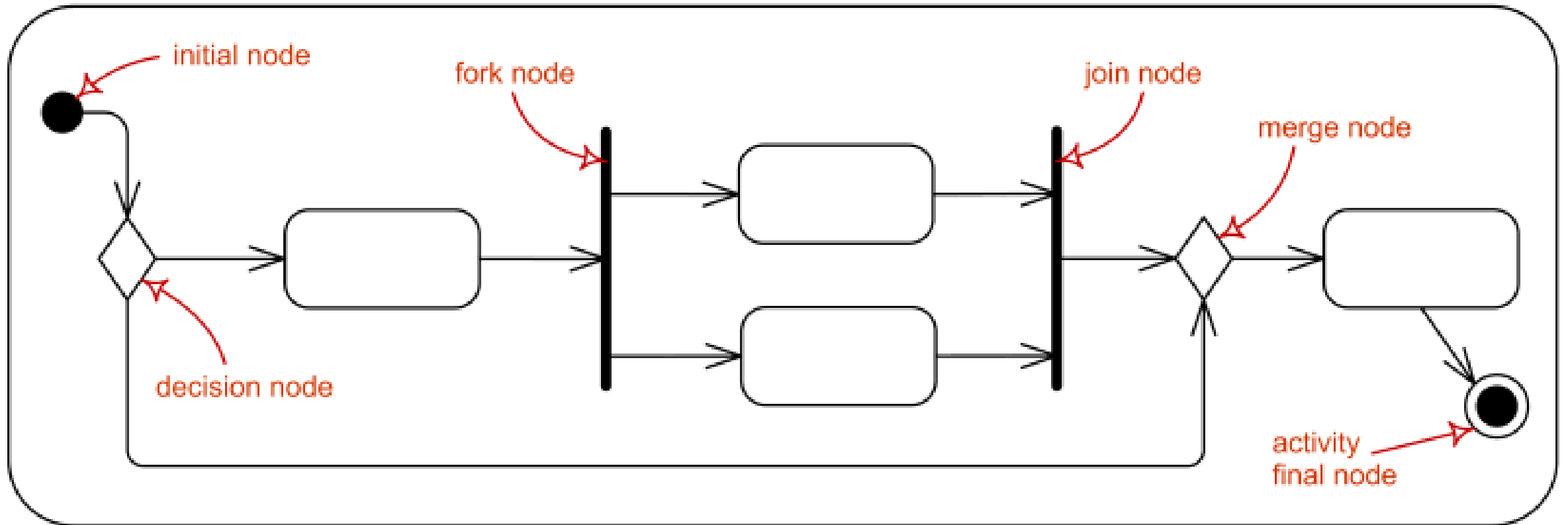
Developing a Software System for a Client

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Activity – 6 : “Order Processing System”

1. Once the order is received by the customer service department, several other departments too operate on various activities to complete the order.
2. The fulfillment department should fill the order (prepare goods to deliver) and deliver the order to customer.
3. Meanwhile, as soon as the order is received, customer service department works on sending an invoice to the customer.
4. The finance department will handle the payments from the customer. The shop operates on credit basis. Therefore, receiving payments is not mandatory prior to the delivery of the goods.
5. Customer service department can close the order only after completing all the above activities.

Activity Diagram Notations



References

- The Unified Modeling Language Reference Manual – James Rumbaugh, Ivar Jacobson, et al
- UML Distilled: A Brief Guide to the Standard Object Modeling Language by [Martin Fowler](#), [Kendall Scott](#)
- Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and the Unified Process by [Craig Larman](#)
- [The Complete UML Training Course, Student Edition](#) by Grady Booch, et al
- UML Explained by [Kendall Scott](#)

Thank You...!!!